

Architecture and Strategy Blueprint for an Autonomous, Self-Improving Hybrid Trading System in Indian F&O Markets

The pursuit of consistent alpha in modern financial markets necessitates the deployment of trading architectures that transcend static rule-based algorithms. An autonomous, self-improving system requires a sophisticated synthesis of deterministic quantitative models, machine learning-driven cognitive agents, resilient infrastructure, and rigorous adherence to evolving market microstructures. Designed to operate across the National Stock Exchange of India (NSE) and the Multi Commodity Exchange (MCX), the target system utilizes a multi-agent topology to continuously rotate the investable universe, execute positional strategies, capture parabolic volatility expansions, and dynamically manage risk.

By amalgamating script-based execution logic with artificial intelligence (AI) evaluation mechanisms, the architecture achieves a continuous feedback loop. This comprehensive blueprint details the infrastructural deployment across AWS Lambda and Oracle Virtual Private Server (VPS) environments, the mathematical foundations of the pricing models, the multi-agent cognitive core, the position-sizing algorithms, and the critical adaptations required to navigate the stringent regulatory framework introduced by the Securities and Exchange Board of India (SEBI) between 2024 and 2026.

1. Topography of the Hybrid Infrastructure

The infrastructural backbone of the trading system relies on a hybrid cloud deployment, strategically partitioning responsibilities between an Oracle VPS and AWS Lambda. This architectural bifurcation optimizes the system for the dual requirements of persistent, low-latency state management and highly scalable, event-driven computation.

1.1 The Oracle VPS: Persistent Stateful Execution and Low-Compute Stack

The Oracle VPS serves as the persistent execution environment and the central nervous system for live market operations. It hosts the OpenAlgo framework, an open-source, self-hosted algorithmic trading platform explicitly designed for Indian markets¹. OpenAlgo bridges the proprietary trading logic with broker APIs, providing an abstraction layer that standardizes requests across 35 different broker plugins, including Fyers and Zerodha, via a unified symbol format³.

To optimize compute costs on the free-tier Oracle VPS and avoid expensive cloud data warehouses, the system leverages the 2026 standard for low-compute quantitative processing: DuckDB and Polars⁵. DuckDB acts as an in-process analytical SQL engine to rapidly query gigabytes of local tick data, while Polars replaces legacy Pandas for hyper-fast, multi-threaded DataFrame manipulation⁶. This drastically reduces CPU overhead and memory constraints, allowing advanced quant logic to run smoothly on minimal hardware.

The Oracle VPS environment is vital for maintaining persistent WebSocket connections and in-memory order books, processes that cannot survive the stateless, ephemeral nature of serverless functions⁷. By utilizing HTTPX connection pooling within OpenAlgo, the Oracle VPS reuses outbound connections to reduce setup overhead, ensuring that execution latency remains strictly minimized during periods of high market volatility⁴.

1.2 AWS Lambda: Ephemeral Compute and Asynchronous Processing

AWS Lambda operates as the asynchronous, event-driven compute engine, invoked for periodic, computationally intensive, or inherently stateless tasks⁷. The deployment of heavy analytical workloads to AWS Lambda ensures that the Oracle VPS avoids CPU monopolization, protecting the latency-sensitive order execution pathways. Lambda functions are orchestrated via Amazon EventBridge, triggering specific microservices and AI evaluation agents at predefined intervals or in response to specific market events⁷. The seamless communication between the Oracle VPS and AWS Lambda is facilitated by secure REST endpoints and a centralized Redis Pub/Sub messaging broker⁸.

2. Core Service Abstractions and Data Ingestion

The system's operational data flow is divided into three distinct background services, each engineered to handle specific data streams, computational requirements, and fault-tolerance protocols.

2.1 Service 1: Live Market Data and Microstructure Polling

Deployed entirely on the Oracle VPS, Service 1 interfaces directly with the broker's data endpoints (e.g., Fyers API v3) via the OpenAlgo core¹. It maintains a continuous WebSocket connection capable of streaming up to 5,000 symbols simultaneously, capturing real-time price ticks, order book updates, and execution confirmations¹¹.

Because broker APIs enforce strict rate limits—such as the SEBI-mandated 10 requests per second, 200 per minute, and 100,000 per day caps—Service 1 utilizes OpenAlgo's built-in rate-limit-aware client³. When polling for Greeks and option chain data via REST API, Service 1 employs a dynamic center-strike polling mechanism, calculating the current At-The-Money (ATM) spot price and exclusively fetching the 25 In-The-Money (ITM) and 25 Out-Of-The-Money (OTM) strikes most relevant to the immediate trading horizon¹³.

2.2 Service 2: Alternative Data, News, and Macroeconomic Ingestion

Service 2 operates episodically on AWS Lambda, triggered periodically to ingest non-technical data. This service aggregates macroeconomic indicators (e.g., US Treasury yields, Dollar Index, India VIX), global market index futures, and financial news sentiment via external APIs. The unstructured news data is processed through an embedded Natural Language Processing (NLP) pipeline within the Lambda function, which extracts sentiment scores and entity relationships. The structured macroeconomic risk scores are subsequently published to the Oracle VPS via the Redis Pub/Sub channel⁸.

2.3 Service 3: Localized Quantitative Computation Engine

Service 3 resides directly on the Oracle VPS alongside Service 1. It utilizes Polars DataFrames to rapidly ingest tick data and QuantLib Python bindings to compute theoretical options pricing,

implied volatility surfaces, and higher-order Greeks in real time⁵. By localizing this compute on the VPS, the system ensures that the trading agents do not suffer from network latency when querying critical risk metrics.

3. The Cognitive Core: Deterministic Scripts vs. AI Agents

To ensure survival during extreme market volatility, the system strictly delineates between time-critical execution and asynchronous evaluation. Processes that require sub-second reaction times (entry/exit triggers, risk management) are built as **Deterministic Scripts**. Processes that require complex reasoning, pattern recognition, and unstructured data parsing are built as **AI Agents**.

Crucially, the system features a **Bidirectional Fallback Architecture**:

- **Agent fails -> Script takes over:** If an AI Agent hits an API timeout, hallucinates, or fails to return a valid structured payload, a secondary deterministic script triggers immediately based on pre-computed quantitative baselines (e.g., Z-scores).
- **Script fails -> Agent takes over:** If a deterministic execution script crashes due to a localized server error, an overarching AI Health Agent detects the frozen process, triggers PagerDuty alerts, and uses a secondary broker API connection to force-liquidate open positions to flat.

Module Designation	Architecture Type	Primary Responsibility	Fallback Mechanism
Execution Script 1	Deterministic Script	Positional Strategy Execution	Hard-coded stop losses at broker level.
Hybrid Agent 2	AI Evaluation + Script Execution	Parabolic Setups & CAS Lottery Trades	Defaults to no-trade if AI fails to parse microstructure.
Risk Script 3	Deterministic Script	Dynamic Trailing Stops & Theta Pre-Exits	Fallback to SEBI mandated margins.
AI Agent 4	Generative AI / NLP	Dynamic Universe Selection & Sector Rotation	Script defaults to previous week's asset universe.
AI Evaluator	Generative AI (LLM)	Daily Post-Market	Script prevents

		System Tuning	hyperparameter shifts > 5%.
--	--	---------------	-----------------------------

3.1 AI Agent 4: Dynamic Universe Selection and Sector Rotation

AI Agent 4 operates on a weekly schedule via AWS Lambda. It programmatically selects the optimal universe of stocks and indices for the upcoming week. First, a deterministic script screens the NSE 500 universe using the National Stock Exchange's strict eligibility rules for derivatives (₹75 lakhs median quarter-sigma order size, ₹75 crores average daily turnover)¹⁷. Once the liquidity baseline is established, the AI parses the macroeconomic data from Service 2 and applies a multifactor model that calculates normalized Z-scores for four fundamental and technical factors: Quality, Value, Momentum, and Low Volatility. The composite score is:

$$\text{Multifactor Score} = (0.35 \times Z_{\text{Quality}}) + (0.25 \times Z_{\text{Value}}) + (0.25 \times Z_{\text{Momentum}}) + (0.15 \times Z_I$$

To prevent overlapping bets, the AI calculates the pairwise Pearson correlation coefficients. If any pair exhibits a correlation greater than 0.50, the asset with the lower historical risk-adjusted return is discarded.

3.2 Execution Script 1: Positional Strategy Baskets

Execution Script 1 is purely deterministic, managing systematic, medium-horizon positional trades. It does not think; it merely executes parameters defined by AI Agent 4.

Strategy Basket	Primary Instrument	Objective & Methodology	Risk Management Profile
Index Delta-Neutral Harvesting	Nifty / BankNifty Options	Captures variance risk premium via Iron Condors and Strangles.	Strict max-loss limits; avoids naked selling overnight to prevent gap risk.
Equities Momentum Breakout	NSE Stock Futures	Capitalizes on volatility expansions in the Agent 4 universe.	Dynamic trailing stops based on Average True Range (ATR).
Commodity Trend-Following	MCX Crude Oil Futures/Options	Rides macroeconomic directional trends in	Extended holding periods; relies on Black-76 pricing for

		energy markets.	option legs ¹⁸ .
Term-Structure Calendar Spreads	Index Options	Exploits differences in implied volatility across expiry weeks.	Pre-expiry unwinding to avoid SEBI expiry-day margin penalties ²⁰ .

When structuring options trades, Script 1 inverts the Black-Scholes formula in QuantLib to find the exact strike price corresponding to a target Delta¹⁶.

3.3 Hybrid Agent 2: Parabolic Setups and CAS Lottery Trades

Hybrid Agent 2 is the aggressive, asymmetric component of the trading system, allocated a strict maximum of 10% of total portfolio capital. It identifies "super confident" setups for naked option buying using a hybrid approach: AI continuously evaluates market depth, but a deterministic script executes the split-second trade.

Strategy A: Microstructure Order Flow Imbalance (OFI) Agent 2 integrates 50-level market depth data to compute the Multi-Level OFI (MLOFI) across the top 10 levels of the order book²². Following models by Cont, Kukanov, and Stoikov, short-term price changes are driven

by the imbalance between supply and demand²⁴. The discrete increment of OFI at event n is:

$$e_n = \mathbf{1}_{\{P_n^b \geq P_{n-1}^b\}} q_n^b - \mathbf{1}_{\{P_n^b \leq P_{n-1}^b\}} q_{n-1}^b - [\mathbf{1}_{\{P_n^a \leq P_{n-1}^a\}} q_n^a - \mathbf{1}_{\{P_n^a \geq P_{n-1}^a\}} q_{n-1}^a]$$

When MLOFI hits extreme statistical percentiles (e.g., massive bid stacking with zero ask replenishment), a microstructural breakout is confirmed, and a script fires a naked ATM call buy to capture the parabolic ignition²².

Strategy B: The CAS Imbalance Lottery Trade

Leveraging the new 2026 SEBI Closing Auction Session (CAS) rules, this data-backed lottery trade capitalizes on the disconnect between the cash and derivatives market at market close.

1. Continuous trading for eligible F&O cash equities halts completely at **3:15 PM**²⁶.
2. However, the respective F&O derivative contracts (futures and options) continue trading continuously until **3:40 PM**²⁶.
3. Between **3:20 PM and 3:30 PM**, the cash market undergoes a blind auction to determine the closing price. Crucially, the exchange broadcasts the Indicative imbalance quantity—a real-time feed showing the unmatched buy or sell pressure at the tentative closing price²⁸.
4. At exactly **3:27 PM** (just before the random CAS window closure between 3:28 PM and 3:30 PM), Agent 2 analyzes this imbalance data using Polars. If there is an overwhelming, mathematically anomalous buy imbalance in the cash stock, Agent 2's script instantly executes a naked Call Option buy in the still-open derivatives segment.
5. When the official cash closing price prints at 3:30 PM with a massive gap-up reflecting the auction imbalance, the derivative option instantly reprices higher. The script

aggressively liquidates the option at 3:35 PM. While the win rate of this specific setup may be lower (e.g., around 20%), the extremely asymmetrical reward-to-risk ratio on a successful imbalance gap ensures long-term positive expectancy.

3.4 Risk Script 3: Dynamic Lifecycle Management

Risk Script 3 serves as the autonomous, deterministic risk manager. It constantly monitors all open positions, adjusting stop-losses and initiating pre-exits based on incoming real-time pricing¹⁴.

For the options portfolio, Script 3's logic is heavily influenced by Theta (Θ) decay curves, which accelerate exponentially as the contract approaches expiration³⁰. Service 3 utilizes QuantLib to continuously project the expected loss of time value over the subsequent hours¹⁶. If Script 3 detects that a directional move has stalled, and the projected Theta decay exceeds the expected value of the position's remaining momentum, it overrides the originating agent and preemptively liquidates the position³³.

4. Quantitative Pricing Models and Derivatives Adaptation

A core requirement is the ability to adapt seamlessly between equity derivatives (NSE) and commodity derivatives (MCX).

4.1 NSE Equity Derivatives: The Black-Scholes-Merton Model

For NSE stock and index options, the underlying asset is the spot price. The system utilizes the classic Black-Scholes-Merton (BSM) process, assuming the asset follows a geometric Brownian motion with a constant drift¹⁶. Service 3 calculates the theoretical fair value and exact Greeks (Delta, Gamma, Theta, Vega, Rho) using the BlackScholesMertonEngine in QuantLib¹⁹.

4.2 MCX Commodity Derivatives: The Black-76 Model

Commodity options in India derive their value from the underlying futures contract, not the physical spot commodity¹⁹. Futures prices inherently incorporate storage costs and the risk-free rate, making them intrinsically forward-looking with zero drift in a risk-neutral measure¹⁹. To accurately price MCX options, Service 3 shifts to the Black-76 model³⁹.

In the Black-76 model, Call (C) and Put (P) prices are:

$$C = e^{-rT} [FN(d_1) - KN(d_2)]$$

$$P = e^{-rT} [KN(-d_2) - FN(-d_1)]$$

Where:

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Within QuantLib, this is achieved by setting the dividend yield equal to the risk-free rate ($b = 0$), neutralizing the drift term associated with spot assets⁴⁰. Failing to use Black-76 for MCX would cause Execution Script 1 to mismanage hedges drastically¹⁹.

5. Capital Allocation and Position Sizing

The architecture governs position sizing using a dynamic implementation of the Kelly Criterion⁴².

5.1 The Fractional Kelly Criterion

The standard Kelly formula dictates the optimal fraction of capital to risk to maximize long-term compound growth⁴²:

$$f^* = \frac{bp - q}{b}$$

Where:

- f^* = The optimal fraction of total capital to wager on the trade.
- p = The historical probability of a winning trade (Win Rate).
- q = The probability of a losing trade ($1 - p$).
- b = The ratio of the average win to the average loss (Reward-to-Risk ratio)⁴⁵.

Because full Kelly sizing introduces unacceptable drawdowns during black swan events, the system employs a "Fractional Kelly" approach (e.g., $0.25 \times f^*$), reducing variance while preserving geometric compounding⁴³. Furthermore, a deterministic hard cap within Risk Script 3 ensures no single trade ever exceeds 2% of total account equity, overriding the Kelly output if necessary¹⁴.

6. Resilience, Self-Improvement, and Systemic Health

6.1 Autonomous Health Management and State Synchronization

An overarching AI Health Agent daemon monitors API rate limits, WebSocket latency, CPU overhead, and Redis memory utilization. If Service 1 detects a disconnected WebSocket, a

deterministic script initiates an exponential backoff algorithm with jitter:

$$\text{Delay} = \min(\text{base} \times 2^{\text{attempt}}, \text{max}) + \text{jitter}$$

During a disconnect, Redis buffers pending messages. Upon reconnection, Service 1 passes its session ID and the last received sequence number to the broker, allowing the server to replay missed messages and seamlessly resume state. If an unrecoverable fault occurs in the execution scripts, the AI Health Agent invokes its fallback protocol: it connects directly to a secondary broker API, forcefully liquidates all open positions, flushes Redis queues, and triggers PagerDuty/Telegram alerts⁴.

6.2 The AI Evaluator: Post-Market Analysis and Tuning

At the conclusion of each trading session, an AWS Lambda function triggers the AI Evaluator agent. Utilizing the Model Context Protocol (MCP), a Large Language Model (LLM) ingests the highly structured JSONL audit logs generated by OpenAlgo alongside historical macroeconomic context³.

The AI conducts a rigorous post-mortem evaluation to tune hyperparameters. For example, it

recalculates the p and b parameters for the Kelly equation based on the rolling statistical window of the last 100 trades⁴³. To prevent the AI from breaking the system, a deterministic script acts as a safeguard, blocking any AI-suggested parameter change that shifts a core metric by more than 5% in a single day.

7. Navigating the SEBI 2026 Regulatory Landscape

SEBI's sweeping 2024–2026 derivatives overhaul introduced profound changes that the system natively navigates⁴⁸.

7.1 Lot Size Revisions and the 50:50 Cash Margin Rule

SEBI increased the minimum contract value for index derivatives to ₹15–20 lakh⁴⁸. Crucially, effective April 1, 2026, at least 50% of the total margin required for any open F&O position must be held in genuine cash⁴⁸. Execution Script 1 verifies that the fractional Kelly allocation never attempts to utilize pledged non-cash margin beyond this 50% threshold. It also accounts for

the 2026 STT tax hikes by incorporating these elevated transaction costs directly into the b parameter of the Kelly equation⁵⁰.

7.2 The Closing Auction Session (CAS) and Exit Mechanics

For standard positional trades, the new 2026 CAS (August 3, 2026) dictates a complete overhaul of exit strategies.

- **Forced Pre-CAS Exits:** Risk Script 3 is hard-coded to auto-square off all intraday cash equities and affected derivatives strictly before 3:12 PM²⁸.
- **Order Type Prohibitions:** During the CAS order entry window, stop-loss orders are canceled by the exchange at 3:15 PM²⁶.
- **Expiry Day Penalty:** SEBI introduced a mandatory 2% Extreme Loss Margin (ELM) penalty

on short options positions held through expiry day⁴⁸. Risk Script 3 unwinds all calendar spreads prior to the day of expiry to avoid this capital lockup²⁰.

8. Conclusion

By deploying a strict division of labor—assigning low-latency execution to deterministic scripts and asynchronous strategy generation to AI agents—the system achieves extreme fault tolerance through bidirectional fallbacks. Utilizing the 2026 standard of DuckDB and Polars allows high-end quantitative evaluation (like Black-Scholes-Merton and Black-76 MCX pricing) to run efficiently on low-compute Oracle VPS infrastructure. Combined with disciplined Fractional Kelly sizing and the aggressive exploitation of SEBI's 2026 CAS imbalances for asymmetric lottery trades, this architecture stands as a highly resilient, self-improving algorithmic entity capable of thriving in modern Indian derivatives markets.

Works cited

1. GitHub - marketcalls/openalgo-docs, <https://github.com/marketcalls/openalgo-docs>
2. GitHub - marketcalls/openalgo: Open Source Algo Trading Platform, <https://github.com/marketcalls/openalgo>
3. Algo Trading, Options Analytics & Execution Platform - OpenAlgo, <https://www.openalgo.in/getting-started>
4. why-to-build-with-openalgo.md - GitHub, <https://github.com/marketcalls/openalgo-docs/blob/main/why-to-build-with-openalgo.md>
5. Polars vs DuckDB in 2026: When To Pick Which | The AI Vibe, <https://theaivibe.org/blog/polars-vs-duckdb-2026-when-to-pick-which>
6. DuckDB, Polars, and the Death of Pandas? - Python in Plain English, <https://python.plainenglish.io/the-python-data-analyst-stack-in-2026-duckdb-polars-and-the-death-of-pandas-790f92b39722>
7. Crypto Trading Bot: Architecture and Roadmap - Vitalii Honchar, <https://www.vitaliihonchar.com/insights/crypto-trading-bot-architecture>
8. WebSockets with FastAPI and Redis: Build a Real-Time Chat App, <https://wittycoder.in/blog/websockets-fastapi-real-time-chat>
9. Solving modern trading platform challenges with cloud computing, <https://www.nagarro.com/en/blog/cloud-computing-solving-modern-trading-platform-challenges>
10. Scalable Real-Time Apps with Python and Redis: Exploring AsyncIO, <https://itnext.io/scalable-real-time-apps-with-python-and-redis-exploring-asyncio-fastapi-and-pub-sub-79b56a9d2b94>
11. Introducing FYERS API Version 3 (V3.0.0): A Major Update for, <https://fyers.in/community/t/introducing-fyers-api-version-3-v3-0-0-a-major-update-for-improved-algo-trading/12193>
12. FYERS Skills | AI Assistant for Algorithmic Trading APIs, <https://fyers.in/skills>
13. Option Chain Data using API - Questions - FYERS Community, <https://fyers.in/community/t/option-chain-data-using-api/21253>

14. AalsiTrader: AI Algo Trading for Zerodha, Dhan & 3 More, <https://aalsitrader.com/>
15. QuantLib Python Tutorial: Getting Started Guide 2026 - Quantt, <https://www.quantt.co.uk/resources/quantlib-python-tutorial>
16. QuantLib in Python: Pricing Vanilla Options for Greeks Calculation, <https://trader-algoritmico.com/blog/quantlib-in-python-pricing-vanilla-options-for-greeks-calculation>
17. Selection Criteria - Equity Derivatives - NSE India, <https://www.nseindia.com/static/products-services/equity-derivatives-selection-criteria>
18. Best Crude Oil Trading Strategy in 2026 - Mudrex Learn, <https://mudrex.com/learn/best-crude-oil-trading-strategy/>
19. Black-76 Monte Carlo in Python: Pricing Commodity Options the, <https://www.zmahmood.com/writing/black-76-monte-carlo-python>
20. SEBI's 2025 F&O Rules: New Option Trading Strategies - Mastertrust, <https://www.mastertrust.co.in/blog/how-sebi-s-new-lot-size-and-margin-rules-for-f-and-o-in-2025-changed-retail-trading-strategies>
21. Option Pricing Engines - QuantLib-Python Documentation, https://quantlib-python-docs.readthedocs.io/en/latest/pricing_engines/options.html
22. Order-Flow Imbalance (OFI) - Emergent Mind, <https://www.emergentmind.com/topics/order-flow-imbalance-ofi-7dff1686-44cf-4cf4-a602-b24df2b7c56e>
23. How Order Book Imbalances Predict Price Moves Before They Happen, <https://medium.com/@thewealthacademyyt/how-order-book-imbalances-predict-price-moves-before-they-happen-crystal-ball-series-part-2-fd9fc66f86a5>
24. The Price Impact of Order Book Events - IDEAS/RePEc, <https://ideas.repec.org/p/arx/papers/1011.6402.html>
25. The price impact of order book events, <https://www.smallake.kr/wp-content/uploads/2016/05/SSRN-id1712822.pdf>
26. SEBI's Closing Auction Session (CAS): Everything You Need to, <https://www.religareonline.com/blog/sebis-closing-auction-session-cas-everything-you-need-to-know-effective-august-3-2026/>
27. Closing Auction Session (CAS) Explained: A Comprehensive Guide, <https://www.sharekhan.com/financial-blog/blogs/closing-auction-session-cas-explained-a-comprehensive-guide>
28. What is SEBI's Closing Auction Session (CAS), and how does it, <https://support.zerodha.com/category/trading-and-markets/trading-faqs/general/articles/closing-auction-session?ref=finshots.in>
29. Closing Auction Session (CAS) - NSE India, <https://www.nseindia.com/static/products-services/closing-auction-session>
30. Option Theta Explained: Time Decay for Beginners | TradingBlock, <https://www.tradingblock.com/blog/option-theta-time-decay>
31. Theta Decay in Options Trading - Groww, <https://groww.in/blog/theta-decay-in-options-trading>
32. The Truth About ODTE Options Time Decay,

- <https://optionalpha.com/blog/0dte-options-time-decay>
33. ODTE Options Explained: What They Are and How To Use Them, <https://alpaca.markets/learn/0dte-options>
 34. Intraday Options Trading Strategies: Scalping, Volatility Breakouts, <https://blog.quantinsti.com/intraday-options-trading-strategies/>
 35. The Black-Scholes model in QuantLib, <https://www.implementingquantlib.com/2023/11/black-scholes.html>
 36. Mastering the Black-Scholes Model with Python - The AI Quant, <https://theaiquant.medium.com/mastering-the-black-scholes-model-with-python-a-comprehensive-guide-to-option-pricing-11af712697b7>
 37. Commodity Options Trading: Black 76, CTM Strikes & Devolvement, <https://zerodha.com/varsity/chapter/commodity-options/>
 38. Black-76 - From First Principles, <https://benjaminwhiteside.com/2021/01/15/black-76/>
 39. Black (1976) Model in Python; Predict European Option Prices on, <https://medium.com/@polanitzer/black-1976-model-in-python-predict-european-option-prices-on-bonds-commodities-and-futures-6bca56c2e6ac>
 40. Option_pricing with python - Kaggle, <https://www.kaggle.com/code/arawn10/option-pricing-with-python>
 41. Black model - Wikipedia, https://en.wikipedia.org/wiki/Black_model
 42. Understanding the Kelly Criterion in Algo-Trading | ALGOGENE, <https://algogene.com/community/post/175>
 43. Kelly Criterion: Position Sizing for Algo Traders - Papers With Backtest, <https://paperswithbacktest.com/course/kelly-criterion-position-sizing>
 44. How to use Kelly Criterion in Trading? - FessorPro, <https://fessorpro.com/blog/kellycriterion>
 45. Commodity Trend Following Strategy Explained | 915 by Groww, <https://groww.in/blog/trend-following-in-commodities>
 46. Kelly Criterion for Stock Trading | Maximize Returns & Manage Risk, <https://groww.in/blog/kelly-criterion-for-stock-trading>
 47. marketcalls/openalgo-webpage: OpenAlgo - Web Portal - GitHub, <https://github.com/marketcalls/openalgo-webpage>
 48. SEBI's New F&O Rules 2024-2026: Full Guide for Retail Traders, <https://aliceblueonline.com/blog/sebis-new-fo-rules-2024-2026-full-guide-for-retail-traders>
 49. SEBI New F&O Rules 2024 to 2026 - Alice Blue, <https://aliceblueonline.com/blog/sebi-new-fo-rules-2024-to-2026>
 50. SEBI F&O Rules 2026: What Has Changed for Options Traders?, <https://www.empiricalacademy.net/blog-details/sebi-fo-rules-2026-what-has-changed-for-options-traders>